Parameters

Server:

- Maximum Budget Q_s
- Period T_s

Task:

- Worst Case Execution Time C_I
- Minimum interarrival Time T_i

Variables

Server:

- Budget c_s
- Server deadline d_s
- Server Utilization/Bandwidth U_s = Q_s/T_s

Tasks:

- Hard Real-Time Task tau_i
- Job J_i
- Arrival time r_i
- Task deadline d_i
- f_i = finishing time of job

Rules

- Server is "active" if there exist served job J_i s.t. r_i <= t <= f_i
  Otherwise server is "idle"
- Served jobs are enqueued in a First in First Out queue

Pseudo Code Soft

```
New Job J_I Arrival:
  If [pending jobs]:
    enqueue new job
  Else if [U_s < c_s / (d_s – r_i)]:
    c_s  = Q_s
    d_s = r_i + T_s
  Start job

Budget exhausted:
  c_s = Q_s
  d_s = d_s + T_s
```

Definitions

Server:

- Budget c_s
- Maximum Budget Q_s
- Period T_s
- Server deadline d_sk

Tasks:

- Hard Real-Time Task tau_i
- Job J_ij
- Arrival time r_ij
- Task deadline d_ij
- Ci = WCET
- Ti = minimum interarrival time

Pseudo Code Soft

```
New Job Arrival:
  If [pending jobs]:
    enqueue new job
  Else if [U_s < c_s / (d_s – r)]:
    c_s  = Q_s
    d_s = r + T_s
  Start job

Budget exhausted:
  c_s = Q_s (replenish)
  d_s = d_s + T_s (reset deadline)
```

Pending Job Example

- Potentially erroneous:
  - Is it okay to approach 0 budget?
  - When 2 tasks arrive exactly at the same time at the server which one is picked?

Pending Job Example – Part 2

Task 1:
Task 2:
Server: 2||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	2
Server Deadline	3
Utilization	?

Pseudo Code Soft

```
New Job J_i Arrival:
   If [pending jobs]:
      enqueue new job
   Else if [U_s < c_s / (d_s – r_i)]:
      c_s  = Q_s
      d_s = r + T_s
   Start job

Budget exhausted:
   c_s = Q_s
   d_s = d_s + T_s
```

Pending Job Example – Part 2

Task 1: ^

Task 2:

Server: 2||

^ = arrival task time delimiter

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Tau_2 1 3

Server Parameter Table

Q_s 2

T_s 3

Server Variable Table

Budget 2

Server Deadline 3

Utilization ?

Pseudo Code Soft

> New Job J_i Arrival:

If [pending jobs]:

enqueue new job

Else if [U_s < c_s / (d_s – r_i)]:

c_s = Q_s

d_s = r + T_s

Start job

Budget exhausted:

c_s = Q_s

d_s = d_s + T_s

Pending Job Example – Part 2

Task 1: ^
Task 2:
Server: 2||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	2
Server Deadline	3
Utilization	0

Pseudo Code Soft

```
New Job J_i Arrival:
  If [pending jobs]:
    enqueue new job
  Else if [U_s < c_s / (d_s – r_i)]:
    c_s  = Q_s
    d_s = r + T_s
  > Start job

Budget exhausted:
  c_s = Q_s
  d_s = d_s + T_s
```

Pending Job Example – Part 2

Task 1: ^
Task 2: ^
Server: 2||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	2
Server Deadline	3
Utilization	0

Pseudo Code Soft

```
> New Job J_i Arrival:
    If [pending jobs]:
        enqueue new job
    Else if [U_s < c_s / (d_s – r_i)]:
        c_s  = Q_s
        d_s = r + T_s
    Start job

Budget exhausted:
    c_s = Q_s
    d_s = d_s + T_s
```

Pending Job Example – Part 2
Task 1: ^
Task 2: ^
Server: 2||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Tau_2 1 3

Server Parameter Table

Q_s 2

T_s 3

Server Variable Table

Budget 2

Server Deadline 3

Utilization 0

Pseudo Code Soft

New Job J_i Arrival:

> If [pending jobs]:

enqueue new job

Else if [U_s < c_s / (d_s – r_i)]:

c_s = Q_s

d_s = r + T_s

Start job

Budget exhausted:

c_s = Q_s

d_s = d_s + T_s

Pending Job Example – Part 2

```
Task 1:     ^
Task 2:     ^
Server:   2||
```

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	2
Server Deadline	3
Utilization	0

Pseudo Code Soft

```
New Job J_i Arrival:
    If [pending jobs]:
        > enqueue new job
    Else if [U_s < c_s / (d_s – r_i)]:
        c_s  = Q_s
        d_s = r + T_s
    Start job

Budget exhausted:
    c_s = Q_s
    d_s = d_s + T_s
```

Pending Job Example – Part 2

Task 1: ^x||
Task 2: ^o||
Server: 2||1||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	1
Server Deadline	3
Utilization	1/3

Pseudo Code Soft

```
New Job J_i Arrival:
    If [pending jobs]:
        enqueue new job
    Else if [U_s < c_s / (d_s – r_i)]:
        c_s  = Q_s
        d_s = r + T_s
    Start job

Budget exhausted:
    c_s = Q_s
    d_s = d_s + T_s
```

Pending Job Example – Part 2

Task 1: ^x|| o ||
Task 2: ^o|| x ||
Server: 2||1||~0||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	2
T_s	3

Server Variable Table

Budget	~0
Server Deadline	3
Utilization	2/3

Pseudo Code Soft

```
New Job J_i Arrival:
  If [pending jobs]:
    enqueue new job
  Else if [U_s < c_s / (d_s – r_i)]:
    c_s  = Q_s
    d_s = r + T_s
  Start job


Budget exhausted:
  c_s = Q_s
  d_s = d_s + T_s
```

Pending Job Example – Part 2

Task 1: ^x|| o ||o
Task 2: ^o|| x ||o
Server: 2||1||~0||2

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Tau_2 1 3

Server Parameter Table

Q_s 2

T_s 3

Server Variable Table

Budget 2

Server Deadline 6

Utilization 2/6

Pseudo Code Soft

```
New Job J_i Arrival:
   If [pending jobs]:
      enqueue new job
   Else if [U_s < c_s / (d_s – r_i)]:
      c_s  = Q_s
      d_s = r + T_s
   Start job

Budget exhausted:
   c_s = Q_s
   d_s = d_s + T_s
```